

Why This Problem

11.8 crore small & marginal farmers (**86% of India's farming community**) feed the nation, yet most still depend on guesswork or shopkeeper advice. This means **low yields, high costs, and debt cycles**.

Just one **wrong fertilizer** choice can burn ₹15,000+ in a season. And with **language + digital barriers**, crores of farmers stay cut off from modern agri-tech.

Imagine: A farmer speaks in **Hindi, Punjabi, Tamil, or Bhojpuri** → and instantly gets field-level, **scientific advice** in his own voice-enabled app.

That's HAL.

Our Solution – HAL

Key Features

- Voice-first app**
Voice-first app and chatbot for low-literate users.
- Soil health**
Soil health and fertilizer recommendations for farmers.
- Pest detection**
Pest and disease detection from photo uploads.
- Weather alerts**
Weather alerts from NASA, ISRO, and OpenMeteo.
- Mandi prices**
Live mandi prices from Government APIs.
- Feedback-driven**
Feedback-driven system that learns and improves over time.

HAL (Harvest Advisory with Linguistic Intelligence) is a **multilingual, AI-powered** advisory system that brings science directly to farmers' fields. By combining **Machine Learning, IoT sensors, satellite data (ISRO + NASA)** and Govt. APIs (**Soil Health Card, Mandi Prices, AI4Bharat**), HAL delivers real-time, hyper-local crop insights in farmers' own languages.

Through a **voice-enabled app and chatbot**, farmers can get **soil health tips, fertilizer guidance, pest detection from photos, weather alerts, and mandi prices** — all simplified into clear, actionable advice they can trust. In short, HAL turns every mobile phone into a smart farming companion.

Expected Outcomes

- 20–30% yield boost** → Imagine wheat farmers producing an extra 4 quintals/acre.
- 15–20% cost savings** → Less chemical overuse, smarter input decisions.
- 30% reduction in fertilizer runoff** → Cleaner soil & water.
- 1 crore+ farmers scalable** → via govt. partnerships & open-source APIs with **Real mandi price tracking**
- Farmers get trust + dignity** → advice in their own language, no middlemen.



What Makes HAL Different (Our Innovation Edge)

- **Space + Soil + Speech = HAL** → **Satellites + IoT + Voice tech** fused into one platform.
- **Hyper-local AI** → Field-level predictions, not just district averages.
- **Voice-first + Multilingual** → Farmers talk, HAL listens.
- **Feedback loop** → HAL evolves with every farmer query.
- **Direct Govt API integration** → **SHC, Mandi, AI4Bharat** → scalable, sustainable.

Cool Factor: HAL isn't just an app, it's a farm companion. Farmers can literally **"talk to their fields"** through HAL.

"When a farmer saves ₹10,000, he feeds his family better. When a village saves lakhs, it changes the economy."